BERNÁT GÁBOR

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SUMMARY

Senior software engineer at Bloomberg operating at staff scope, remote from Los Angeles since 2022. Owns architecture and end-to-end delivery of distributed backend platforms, developer APIs, and enterprise security infrastructure across 13+ organizations: artifact management, enterprise code search, static analysis, and secret scanning. 14+ years building developer tooling used by millions of Python engineers worldwide. Maintains 30+ Python ecosystem projects including virtualenv, tox, build, pipx, and pipdeptree. Python Software Foundation Fellow (2021). Delivered 15+ talks and workshops at PyCon US, EuroPython, PyTexas, and PyLondinium (2018–2026) on packaging standards, supply chain security, virtual environments, Python–Rust interoperability, and AI-assisted development.

EXPERIENCE

Bloomberg — *2025 April – present*
Senior Software Engineer – Developer Experience — *Los Angeles, USA (remote)*

Owns architecture and end-to-end delivery of backend platform services used daily by Bloomberg's 9,000+ engineers across 13+ organizations. Designed the artifact repository portal and its REST APIs for Python package and Docker image discovery: namespace browsing, wheel compatibility filtering, self-service lifecycle management. Improved cache hit rate by 25%. Built event-streaming notification infrastructure on Kafka so downstream services react to artifact changes in real time.

Replaced the legacy Lua-based authentication layer of enterprise code search with a Go service implementing JWT validation, PKCE OAuth2, and session management. Sunset the platform's open-access tier. Migrated source ingestion to S3 object storage, parallelized distributed indexing, and led incident response for a silent data-loss bug that had dropped 34,591 projects from the search index. Instrumented end-to-end observability across all data sources.

Rolled out secret-scanning infrastructure across Bloomberg. Led the cross-org CPython 3.14 migration across 20+ services. Owns linter integrations across the company's static analysis ecosystem and built developer-experience tooling for AI usage tracking. Cut database query volume by 85% for the internal conference platform. Chairs the Python Guild's Packaging Working Group; spoke at PyTexas 2025 and PyCon US 2025, and led a solo workshop on AI-assisted Python development at PyTexas 2026.

Bloomberg — *2022 June – 2025 April*
Senior Software Engineer – Data Technologies, Quality Control — *Los Angeles, USA (remote)*

Built the distributed data-quality platform: metrics ingestion, deduplication, workflow orchestration. Designed and shipped a self-service storage-platform management portal with authentication and configuration APIs. Standardized SSO across multiple services through a shared integration library. Led an accessibility overhaul of the Guild Week conference website with keyboard navigation and screen-reader support. Built the internal Python package registry frontend and conference platform.

Bloomberg — *2016 April–2022 June*
Software Engineer → Senior (2019) – Data Technologies, Quality Control — *London, United Kingdom*

Core developer of the Business Rule Engine, a Python framework that lets financial analysts author validation rules over a microservice platform with automated provenance and quality tracking. Chaired the Python Guild and led its Conference Working Group (2017–2022).

Gravity Research & Development *2011 September–2016 March*
Software and Integration Engineer *Budapest, Hungary*

Built a recommendation platform serving hundreds of millions of daily recommendations (SaaS) for clients such as dailymotion.com and allegro.pl. Handled client integrations, A/B testing, and mentoring.

OPEN SOURCE

Open source software developer *2015–present*
bernat.tech/about

Author and primary maintainer of 30+ Python open source projects. Rewrote virtualenv and tox from scratch as lead maintainer; both rank among the most-downloaded developer tools on PyPI. Led the virtualenv security hardening program and shipped 20+ tox features in 2026 including PEP 751 lock-file support and a JSON Schema for IDE integration. Took over build as primary maintainer and brought it to 100% test coverage. Revitalized pipx and modernized pipdeptree. Extended filelock for NFS/HPC and async environments. Contributed PEP 751 lock-file support to uv. Also maintains platformdirs, sphinx-autodoc-typehints, pyproject-fmt, tox-uv, and pytest-memray. Drove security hardening across 20+ ecosystem repositories. PSF Fellow (2021 Q3). Blog post series on Python type hints reached #1 on Hacker News.

EDUCATION

Budapest University of Technology and Economics, Hungary. *2011–2013*
Master's degree in Engineering Information Technologist (grade 4.57/5).

Sapientia – Hungarian University of Transylvania, Romania. *2007–2011*
Bachelor's degree in Computer Science (grade 9.70/10).

TECHNICAL STRENGTHS

Languages	*Python* (primary), Go, TypeScript, Rust, Java, Kotlin, C/C++
Frameworks	FastAPI, React, Flask
Infrastructure	Docker, Kubernetes, Redis, Kafka, PostgreSQL, S3, HAProxy, GitHub Actions, Jenkins
Standards	PEP 751, OAuth2/OIDC, JWT, PKCE, JSON Schema
Tools	git, vim, tox, uv, ruff, ty, VS Code, IntelliJ IDEA
Specialties	Developer platforms, API design, packaging, static analysis, supply-chain security, distributed systems